

Contents

1	Intro	2
2	Year: 201516, Magnet: Tot	3
2.1	Selection: <code>bdtGraNEW</code> > 0.86	3
2.1.1	Final state: <code>PIPI</code>	3
2.1.2	Final state: <code>KK</code>	5
2.1.3	Final state: <code>KPI</code>	7
3	Year: 2017s29r2p2, Magnet: Tot	9
3.1	Selection: <code>bdtGraNEW</code> > 0.86	9
3.1.1	Final state: <code>PIPI</code>	9
3.1.2	Final state: <code>KK</code>	11
3.1.3	Final state: <code>KPI</code>	13
4	Year: 2018, Magnet: Tot	15
4.1	Selection: <code>bdtGraNEW</code> > 0.86	15
4.1.1	Final state: <code>PIPI</code>	15
4.1.2	Final state: <code>KK</code>	17
4.1.3	Final state: <code>KPI</code>	19
5	Year: Tot, Magnet: Tot	21
5.1	Selection: <code>bdtGraNEW</code> > 0.86	21
5.1.1	Final state: <code>PIPI</code>	21
5.1.2	Final state: <code>KK</code>	23
5.1.3	Final state: <code>KPI</code>	25
6	Notes	27

1 Intro

2 Year: 201516, Magnet: Tot

2.1 Selection: $\text{bdtGraNEW} > 0.86$

2.1.1 Final state: PIPI

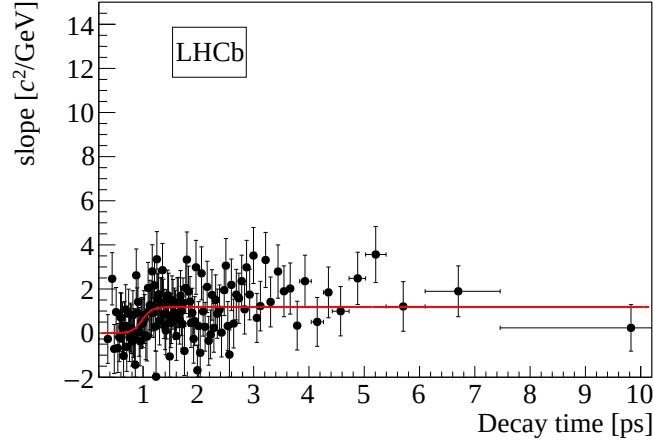


Figure 1: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging: None

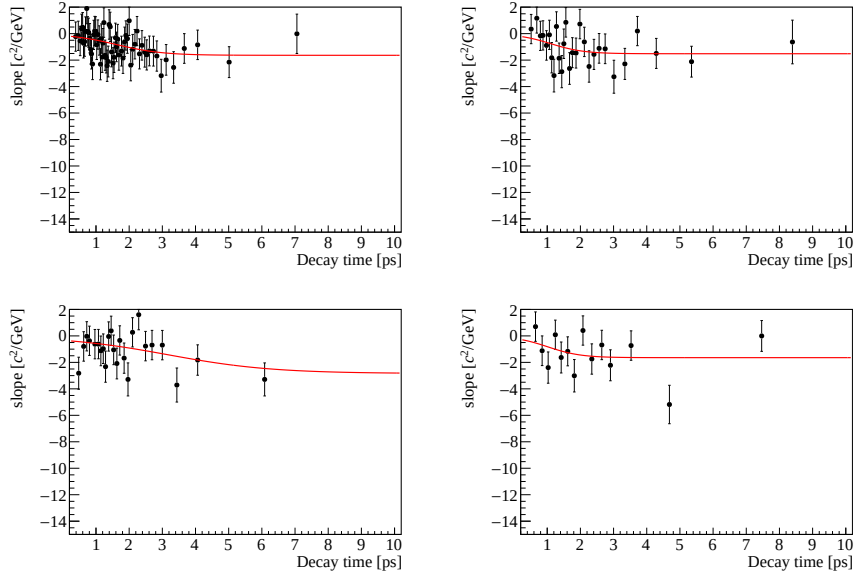


Figure 2: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

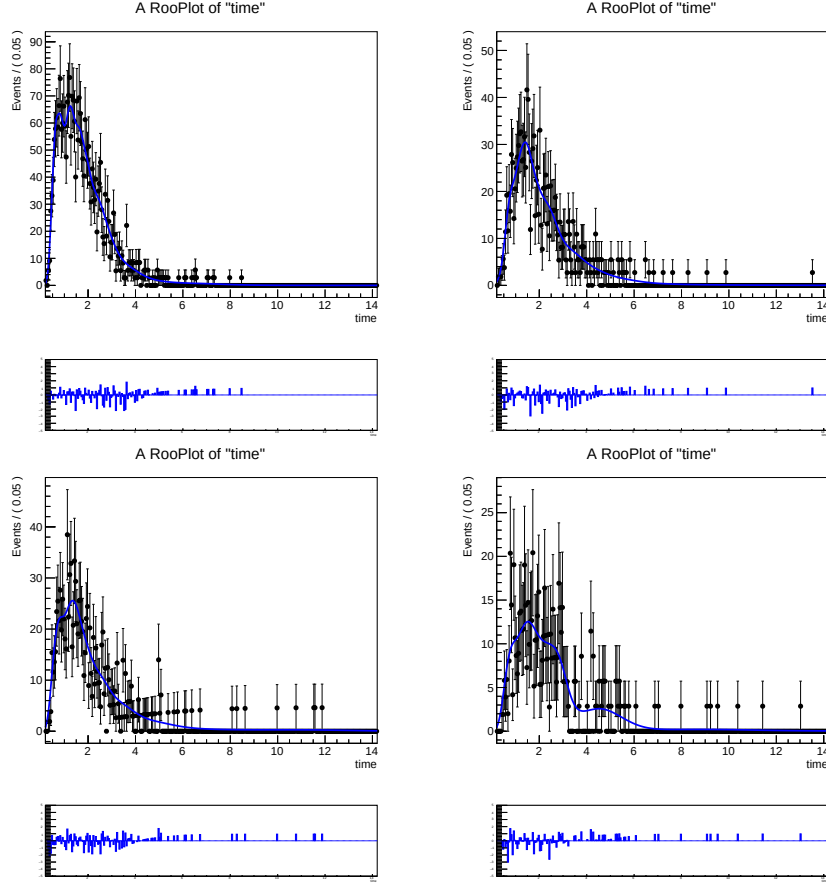


Figure 3: Template for combinatorial background. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

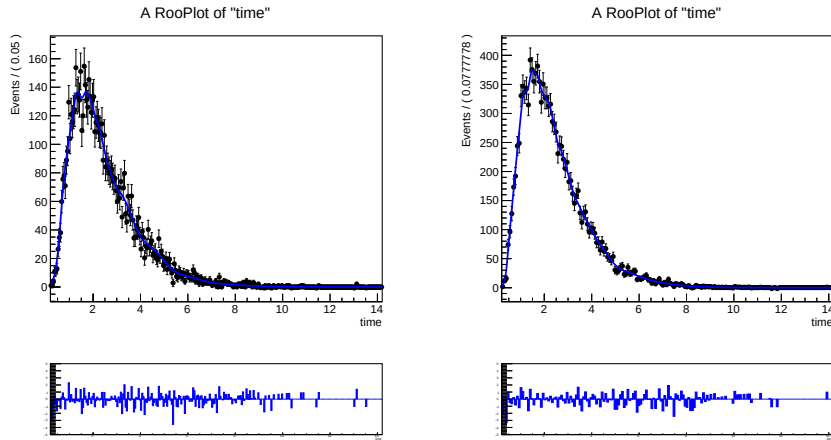


Figure 4: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI.

2.1.2 Final state: KK

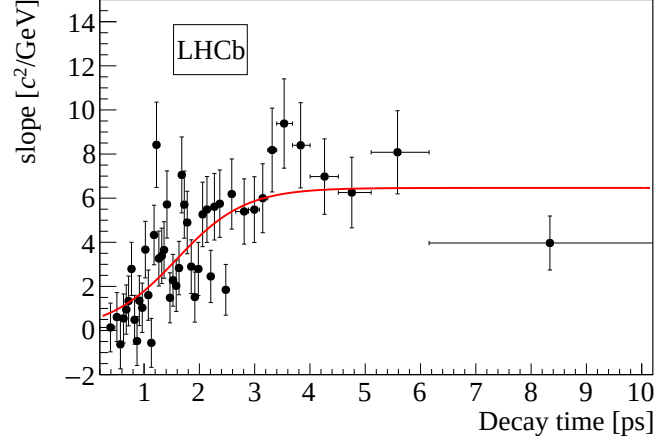


Figure 5: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging: None

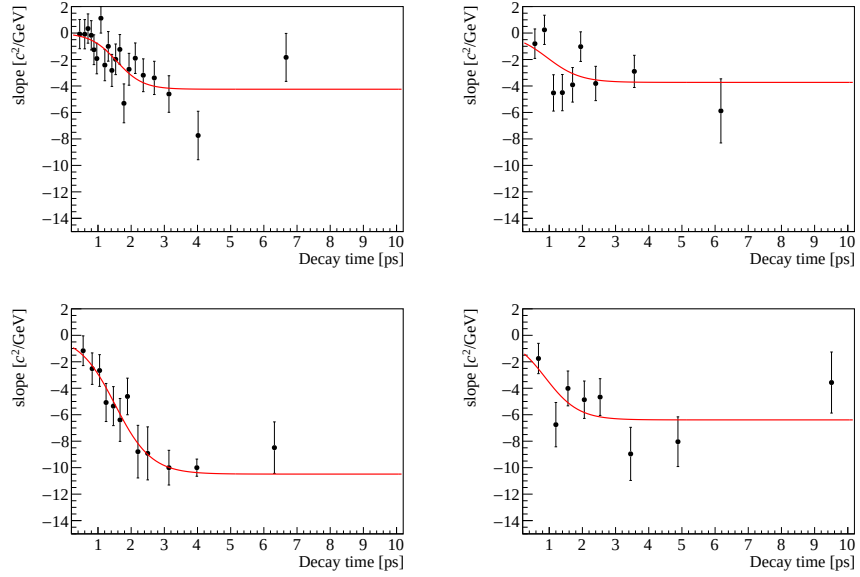


Figure 6: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

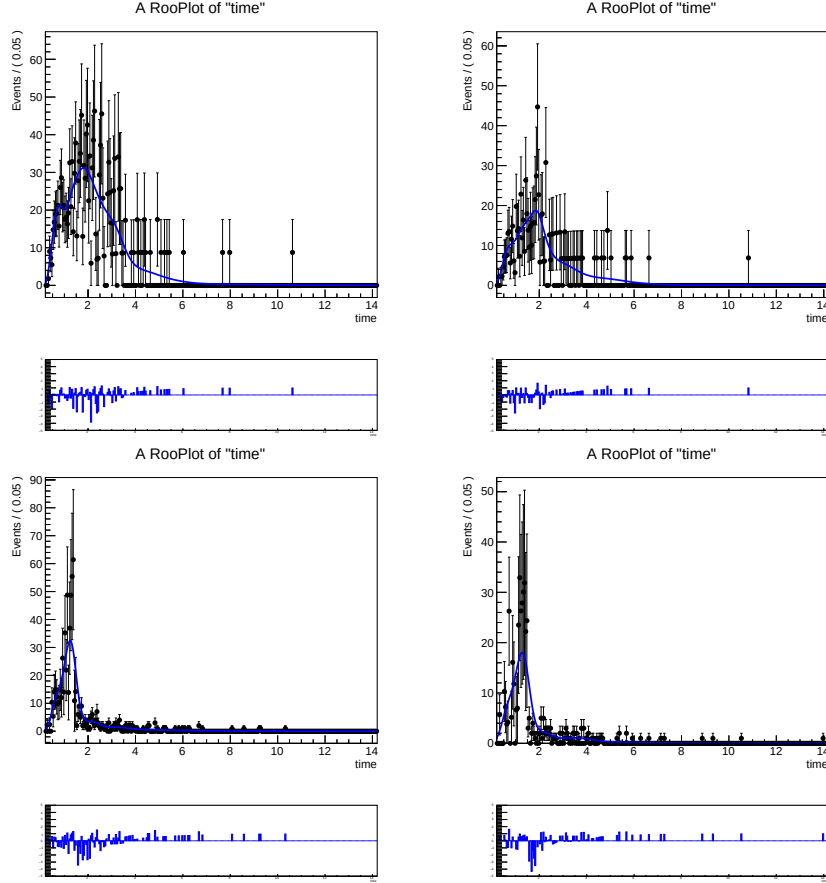


Figure 7: Template for combinatorial background. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

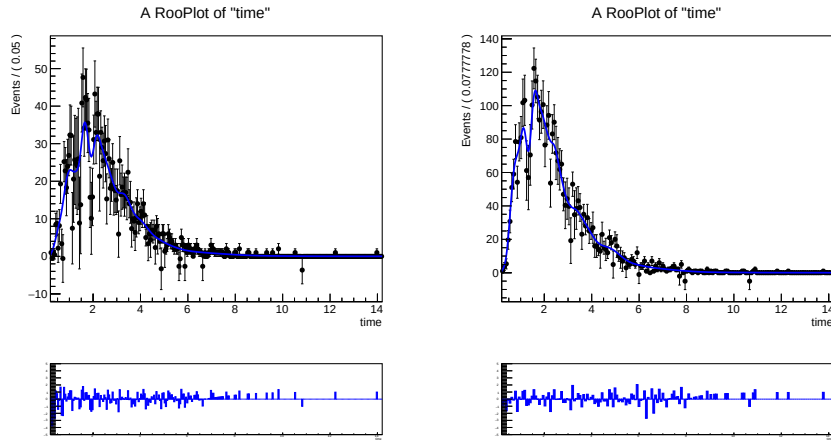


Figure 8: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK.

2.1.3 Final state: KPI

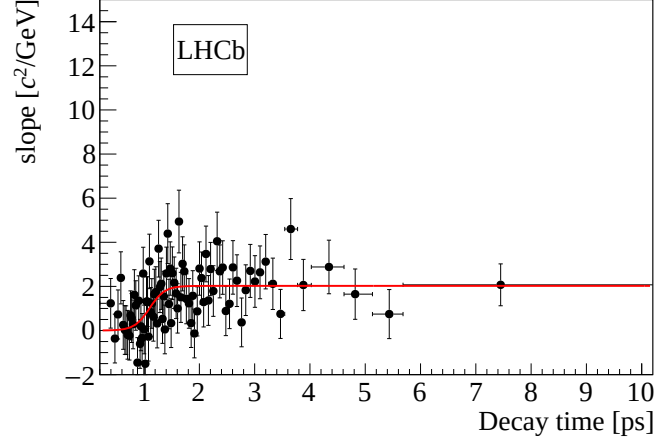


Figure 9: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging: None

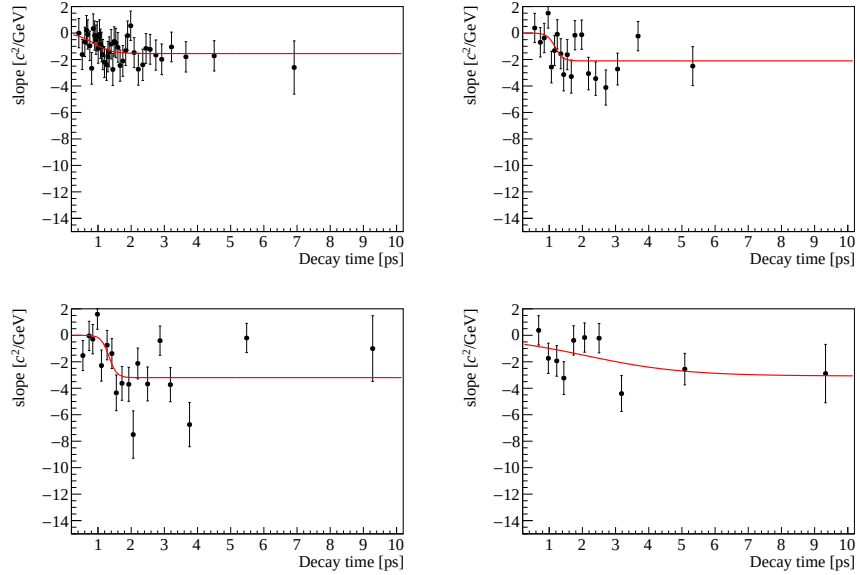


Figure 10: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 201516; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

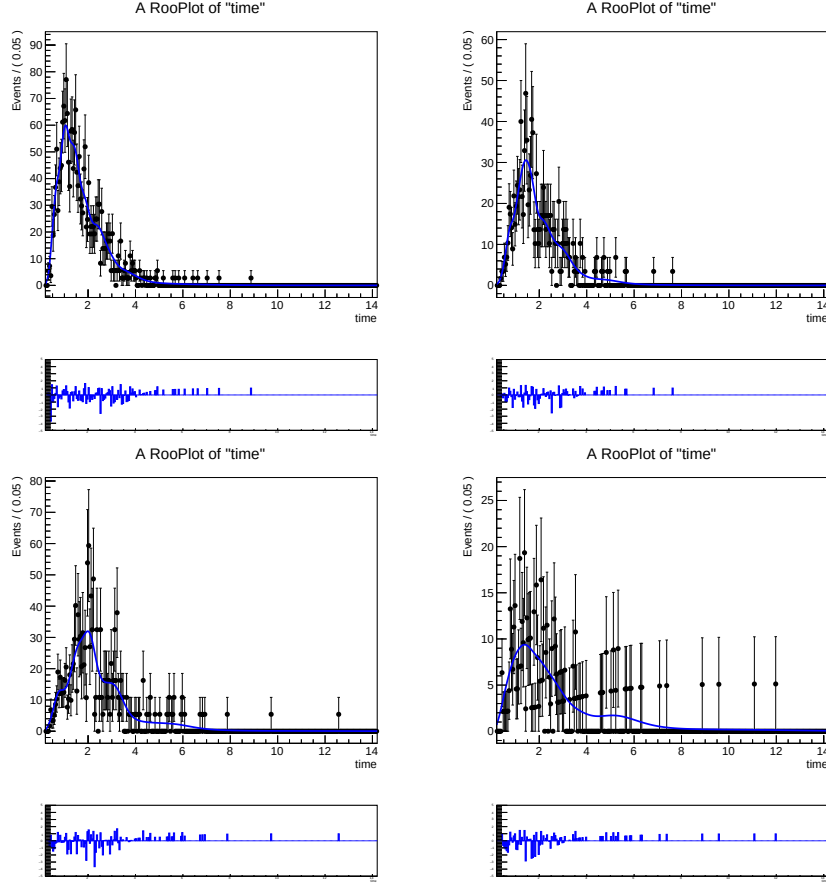


Figure 11: Template for combinatorial background. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

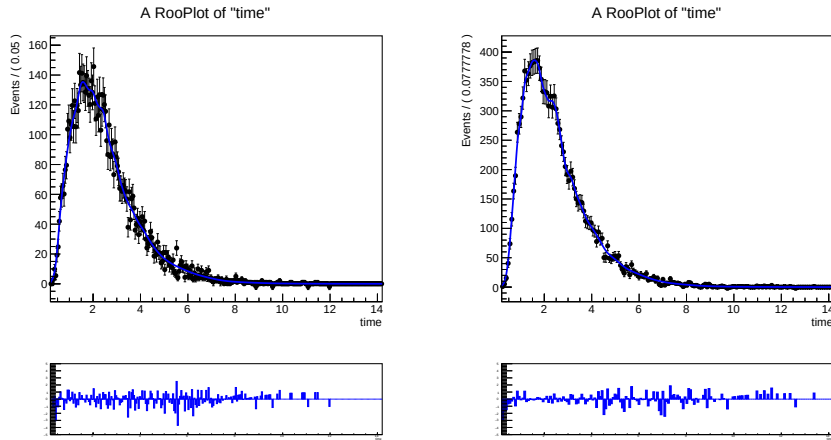


Figure 12: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 201516; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI.

3 Year: 2017s29r2p2, Magnet: Tot

3.1 Selection: $\text{bdtGraNEW} > 0.86$

3.1.1 Final state: PIPI

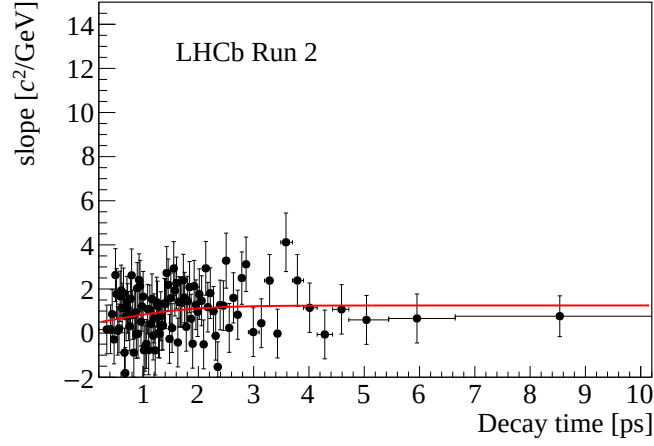


Figure 13: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging: None

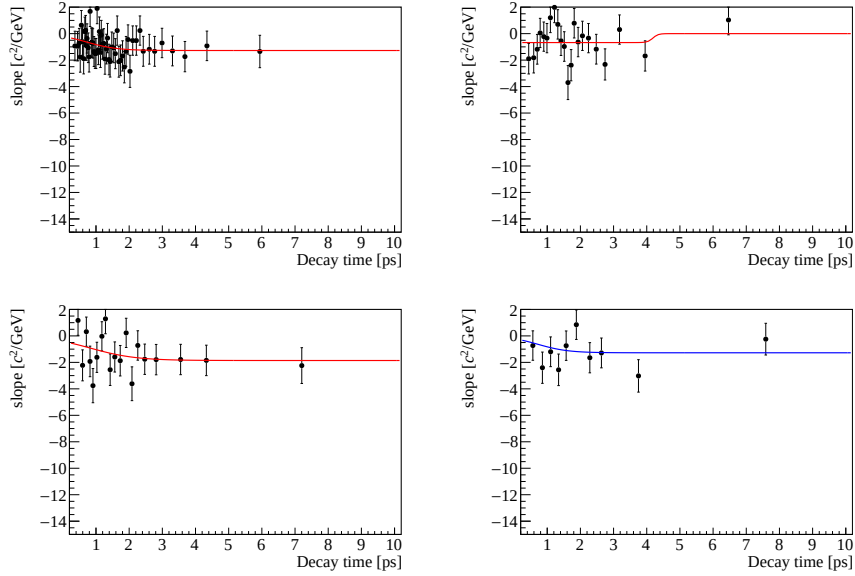


Figure 14: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

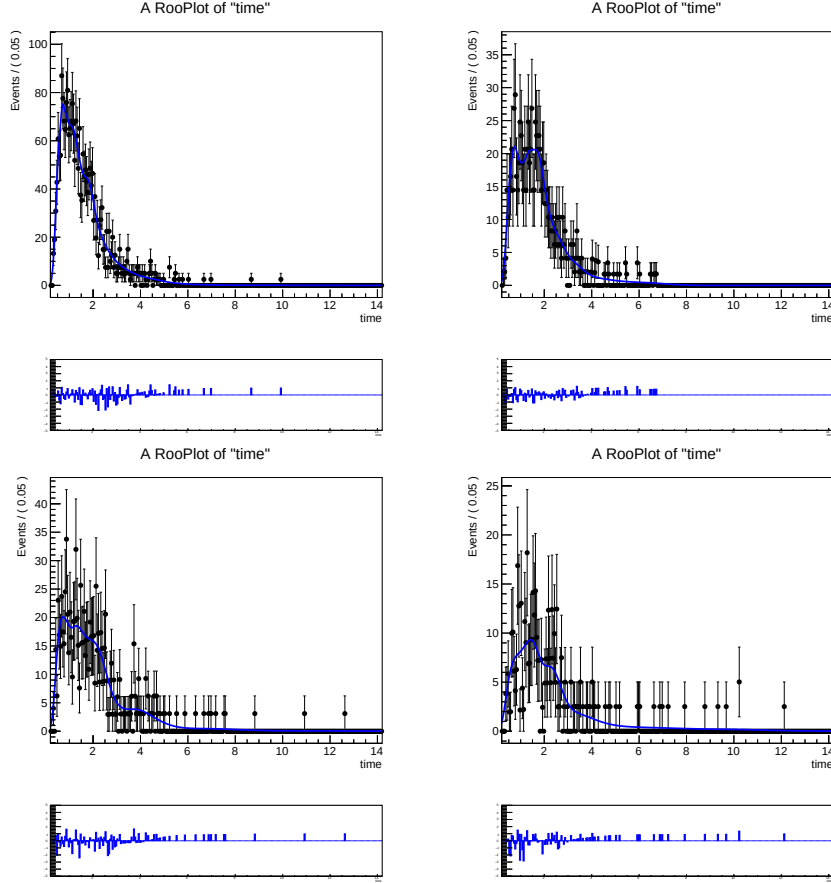


Figure 15: Template for combinatorial background. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

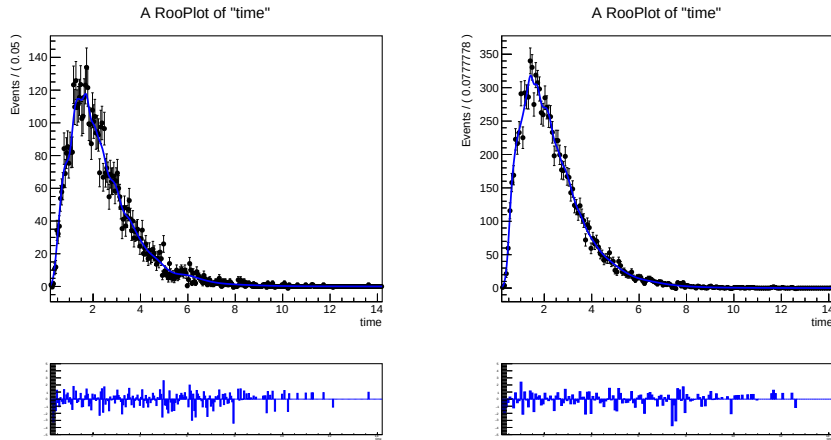


Figure 16: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI.

3.1.2 Final state: KK

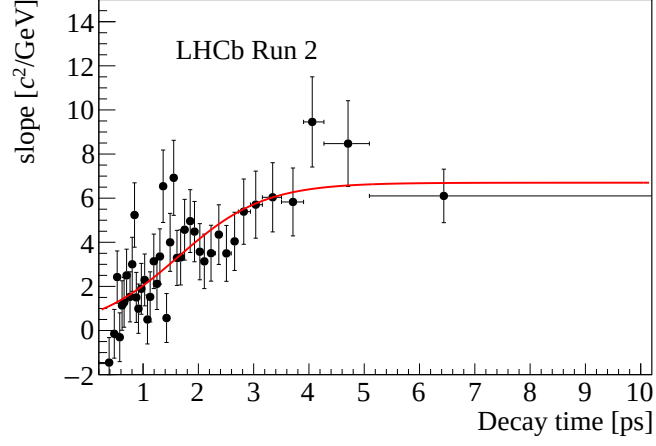


Figure 17: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging: None

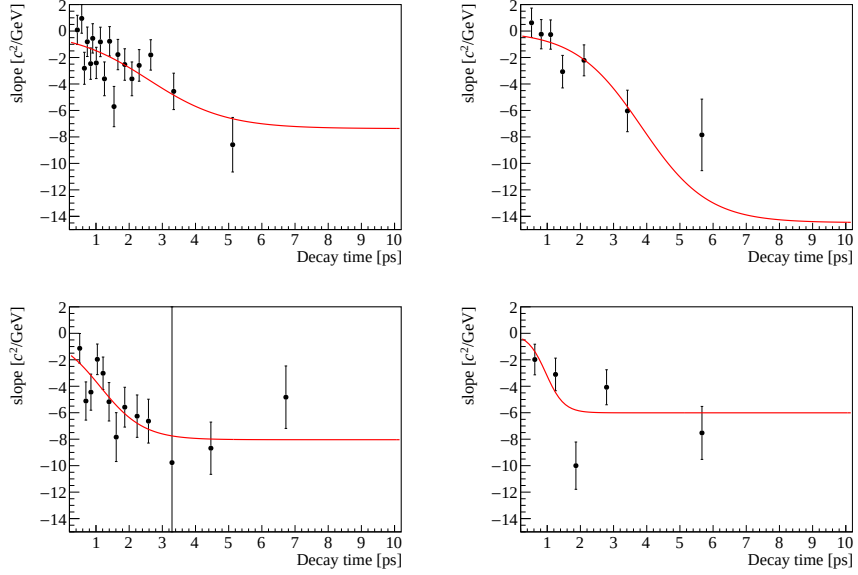


Figure 18: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

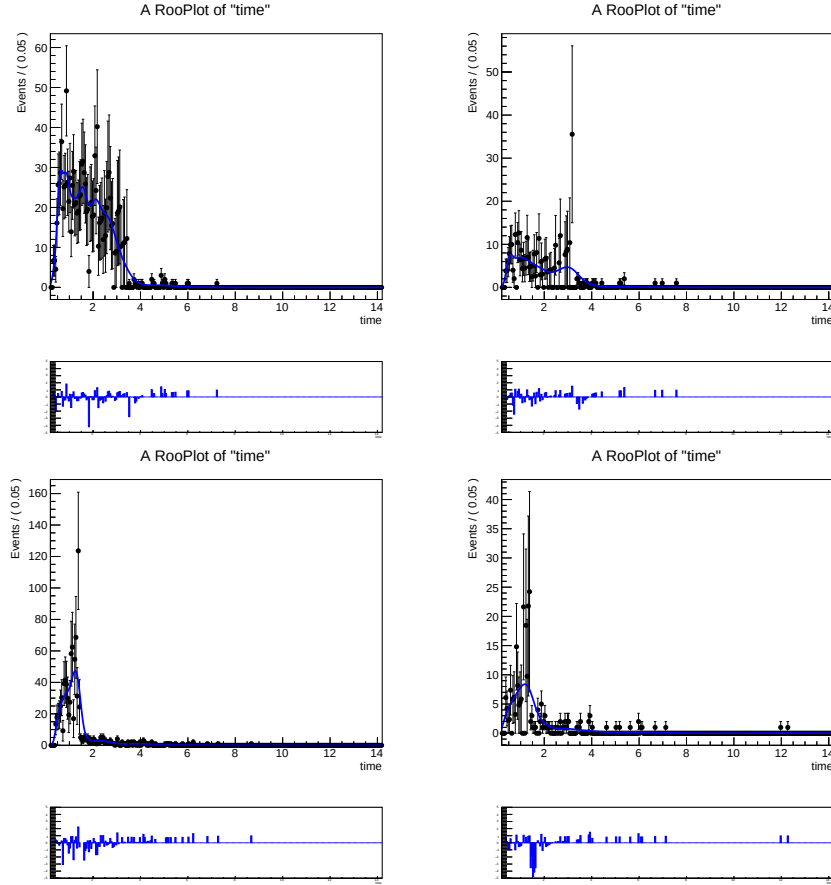


Figure 19: Template for combinatorial background. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

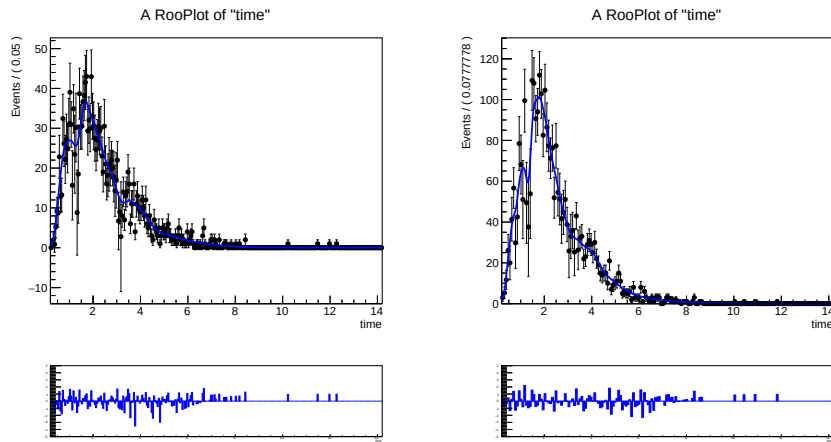


Figure 20: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK.

3.1.3 Final state: KPI

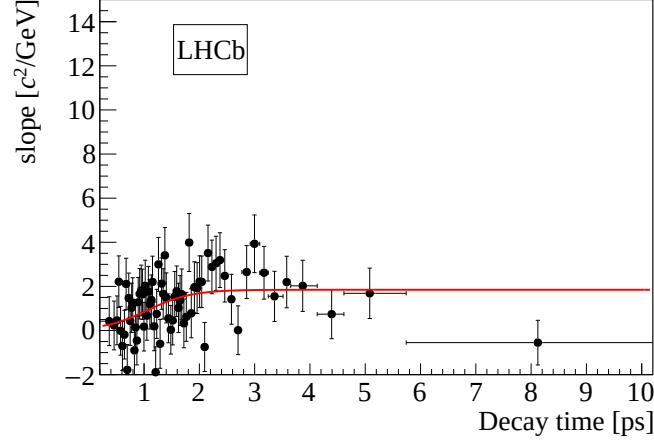


Figure 21: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging: None

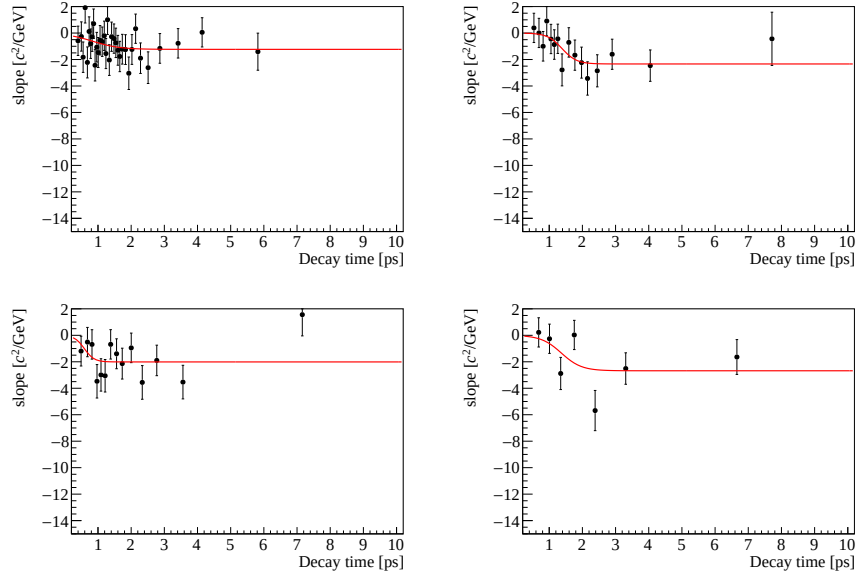


Figure 22: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2017s29r2p2; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

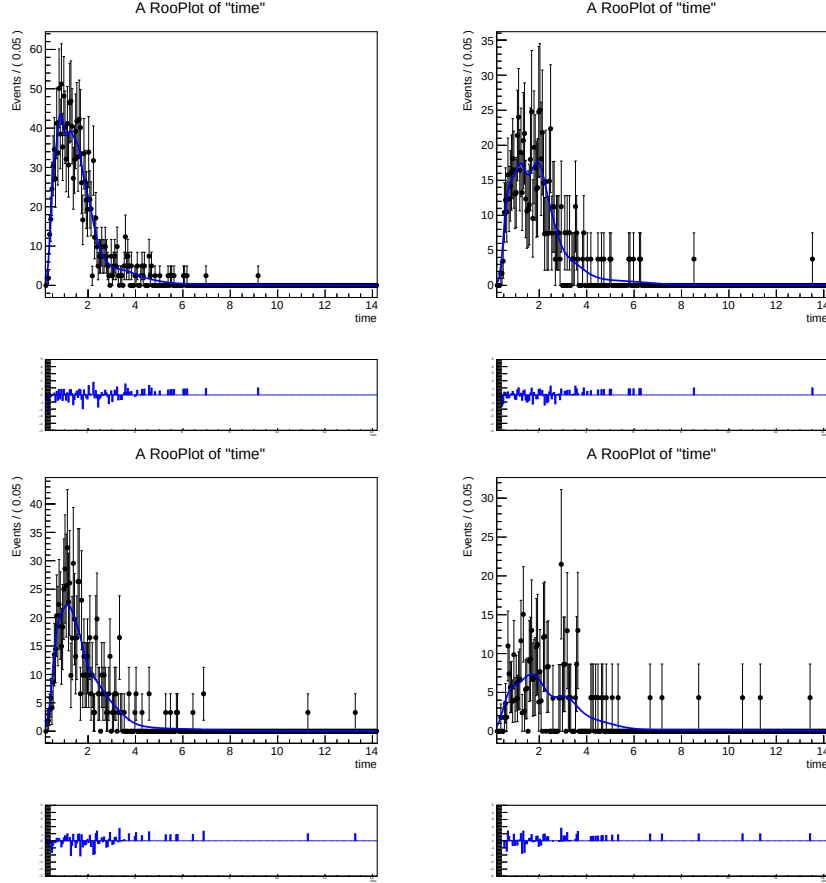


Figure 23: Template for combinatorial background. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

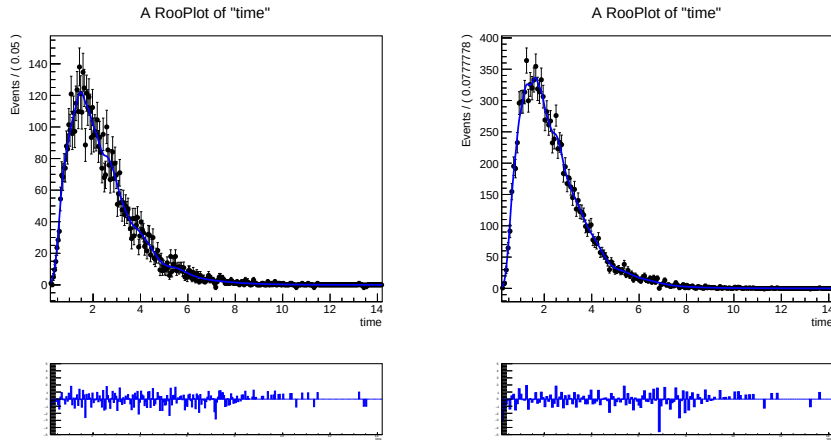


Figure 24: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2017s29r2p2; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI.

4 Year: 2018, Magnet: Tot

4.1 Selection: $\text{bdtGraNEW} > 0.86$

4.1.1 Final state: PIPI

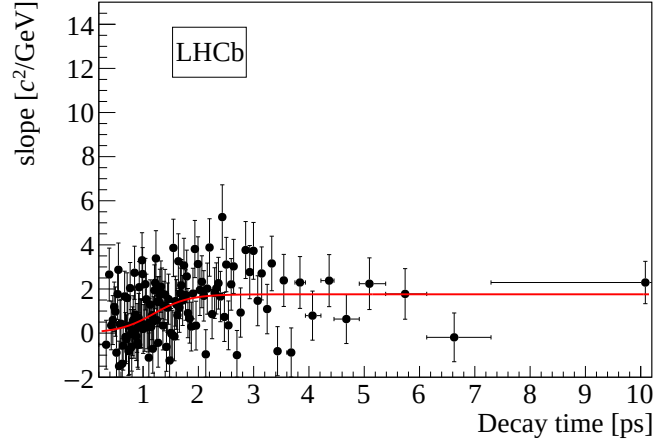


Figure 25: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging: None

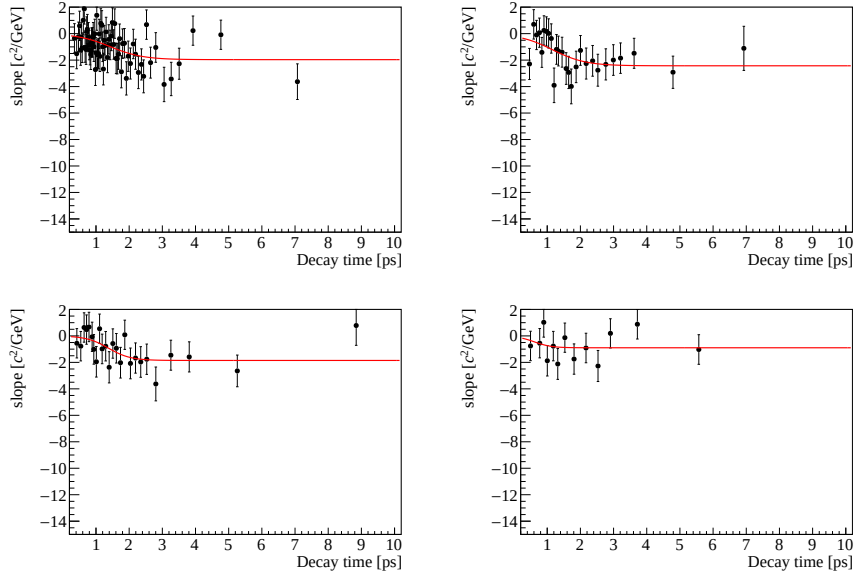


Figure 26: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

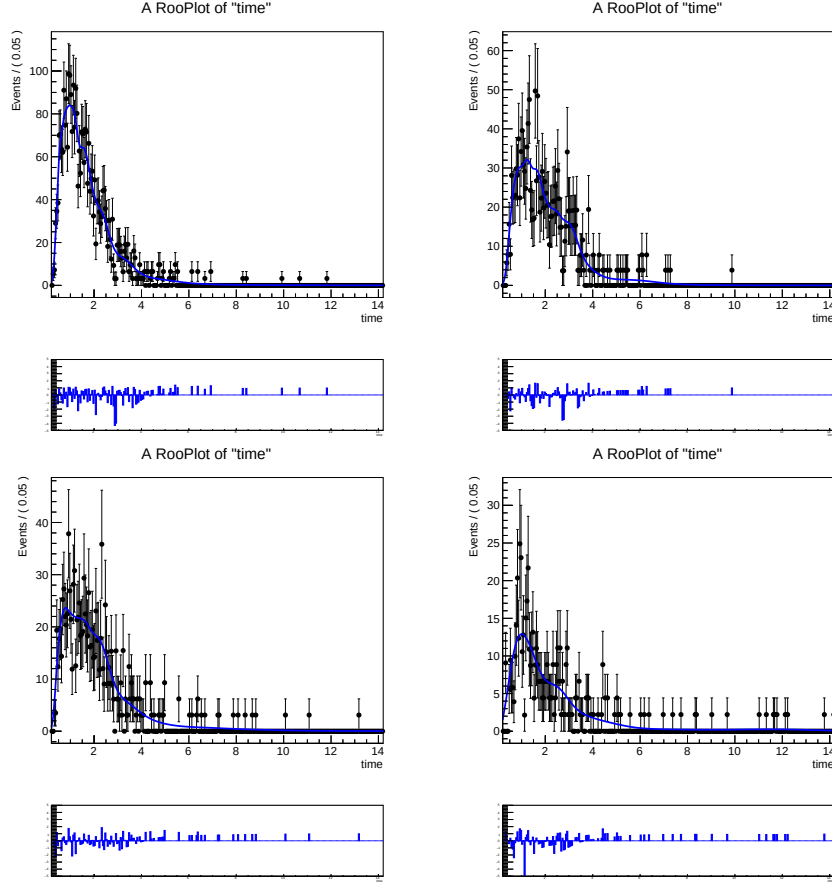


Figure 27: Template for combinatorial background. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPi; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

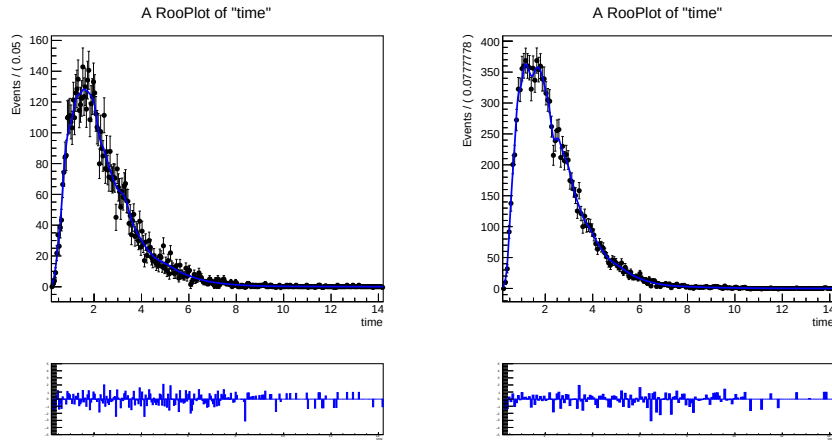


Figure 28: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPi.

4.1.2 Final state: KK

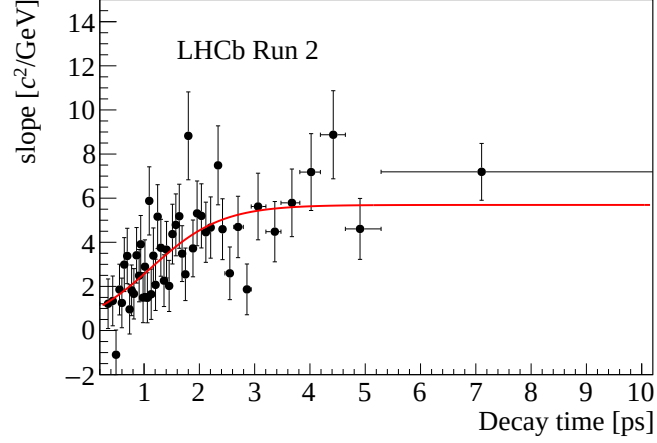


Figure 29: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging: None

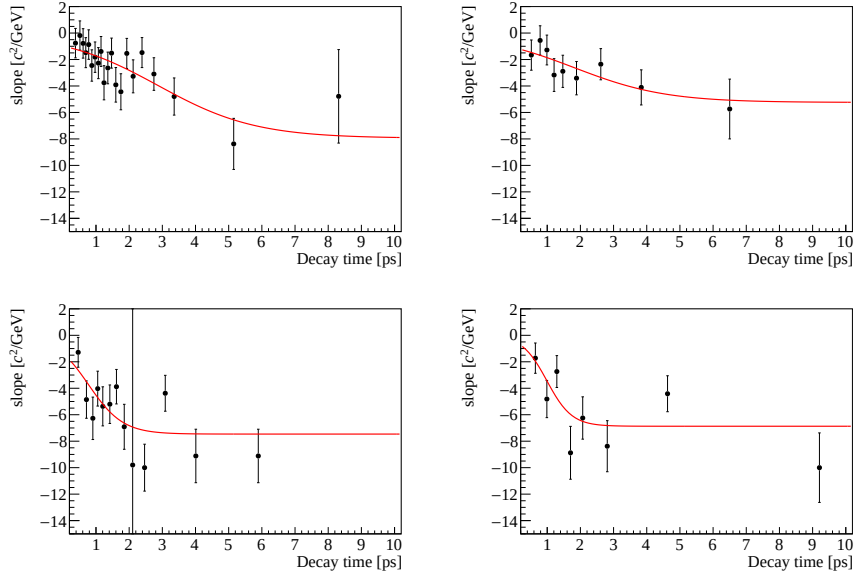


Figure 30: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

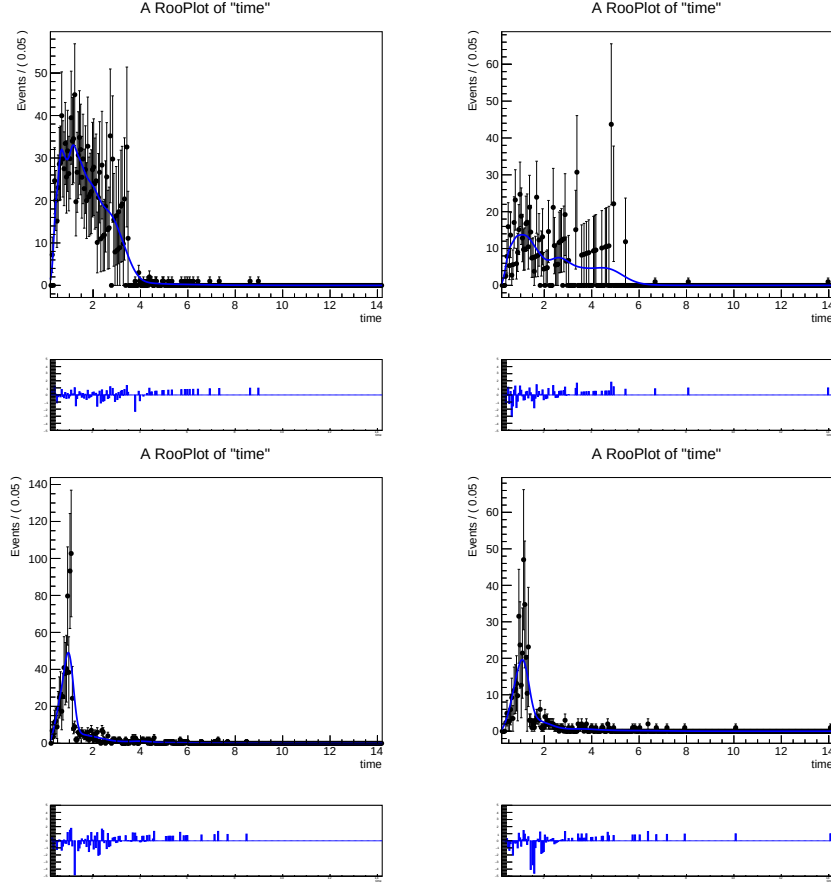


Figure 31: Template for combinatorial background. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

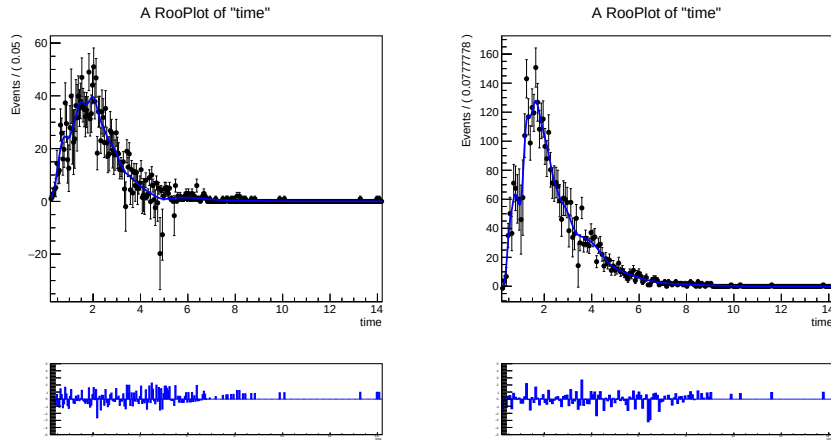


Figure 32: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK.

4.1.3 Final state: KPI

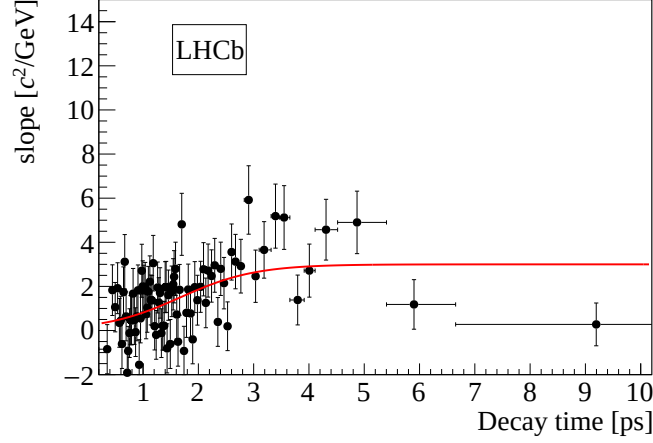


Figure 33: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging: None

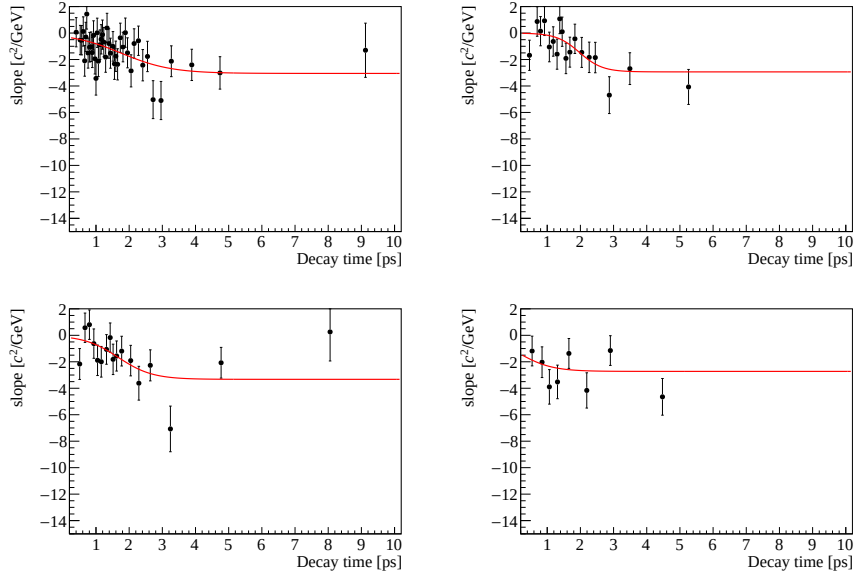


Figure 34: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: 2018; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

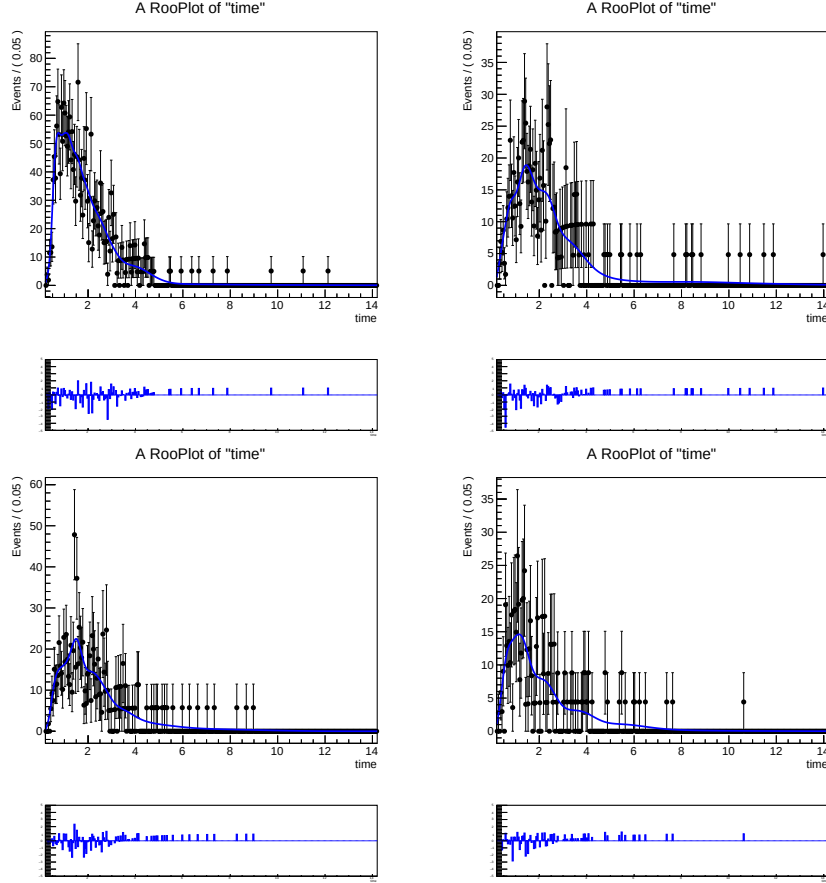


Figure 35: Template for combinatorial background. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

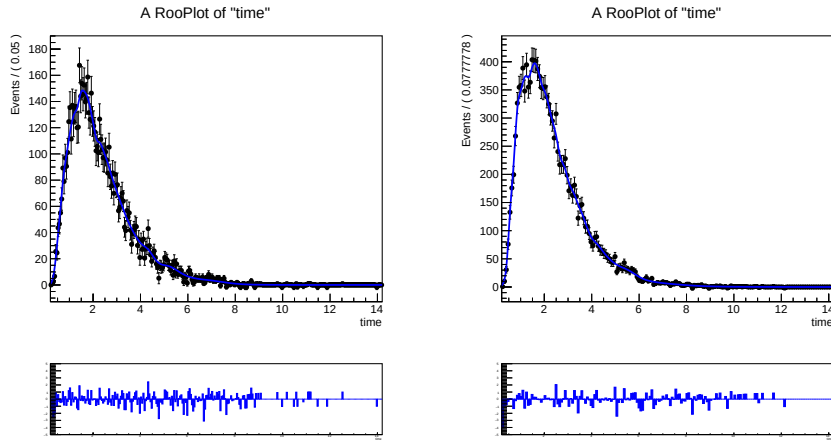


Figure 36: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: 2018; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI.

5 Year: Tot, Magnet: Tot

5.1 Selection: $\text{bdtGraNEW} > 0.86$

5.1.1 Final state: PIPI

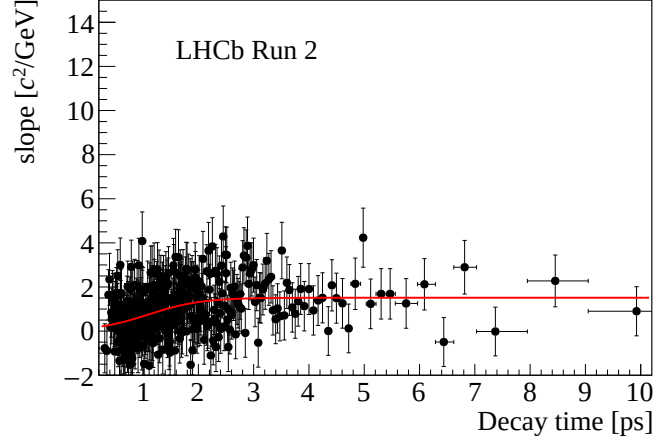


Figure 37: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging: None

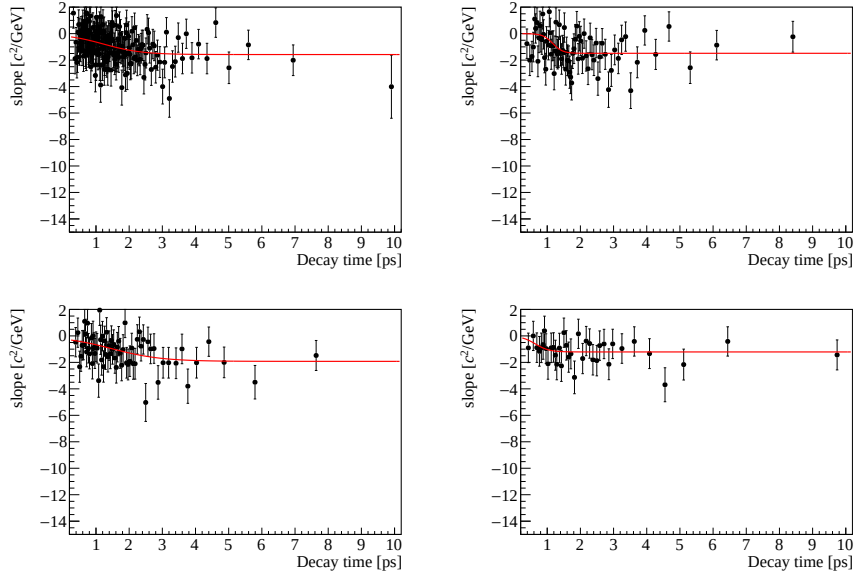


Figure 38: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPI ; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

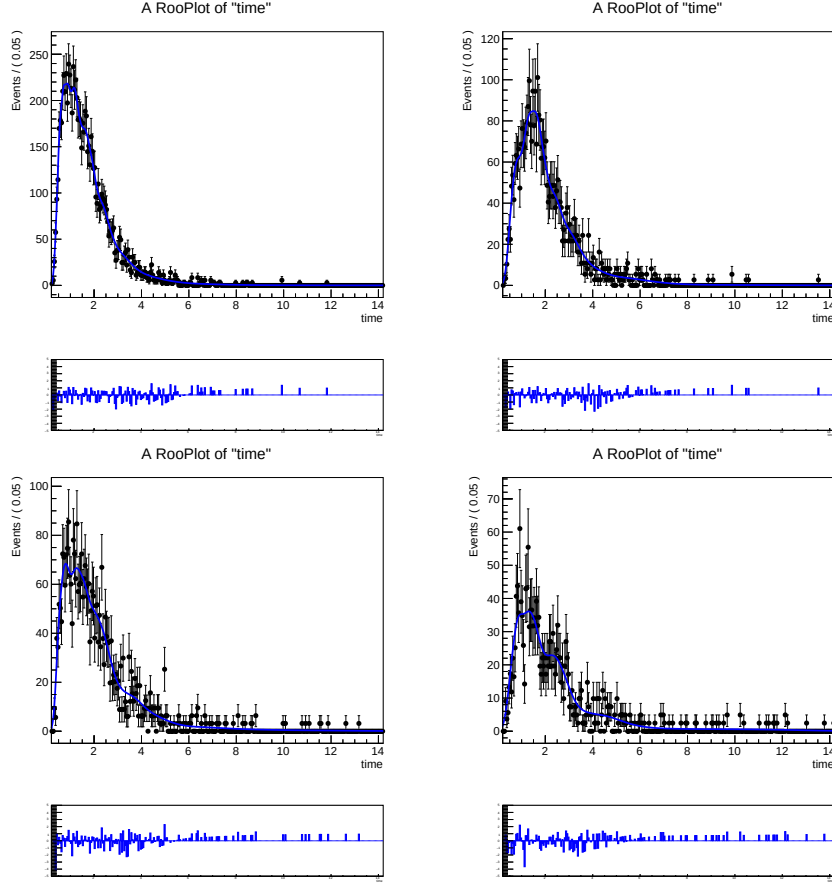


Figure 39: Template for combinatorial background. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPi; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

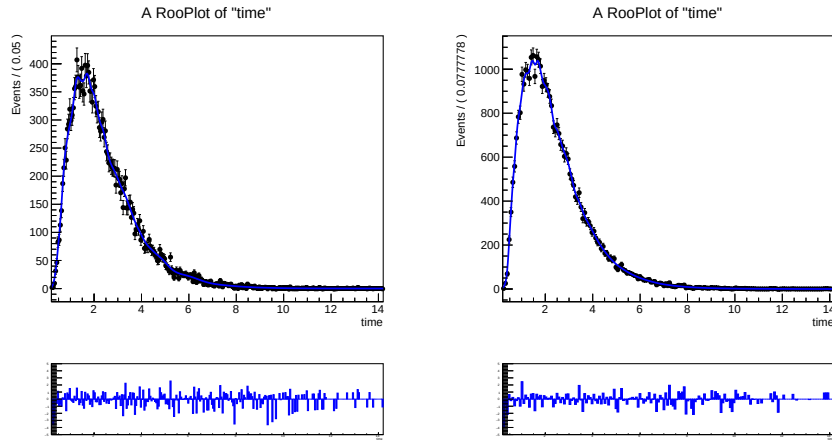


Figure 40: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: PIPi.

5.1.2 Final state: KK

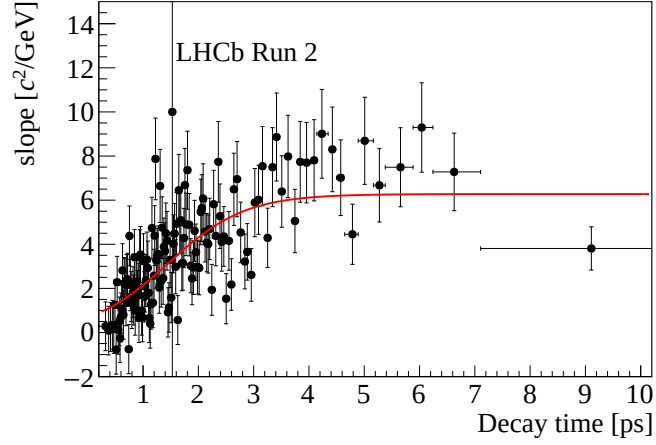


Figure 41: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging: None

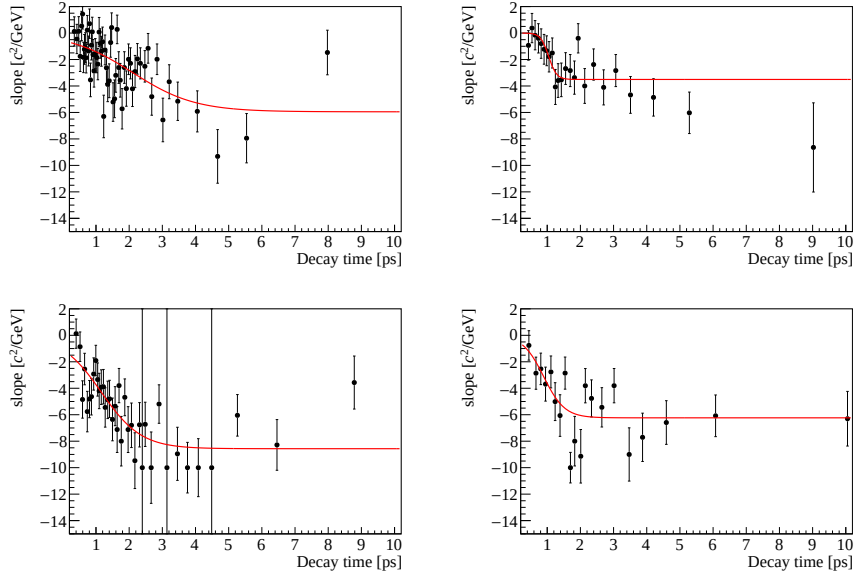


Figure 42: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

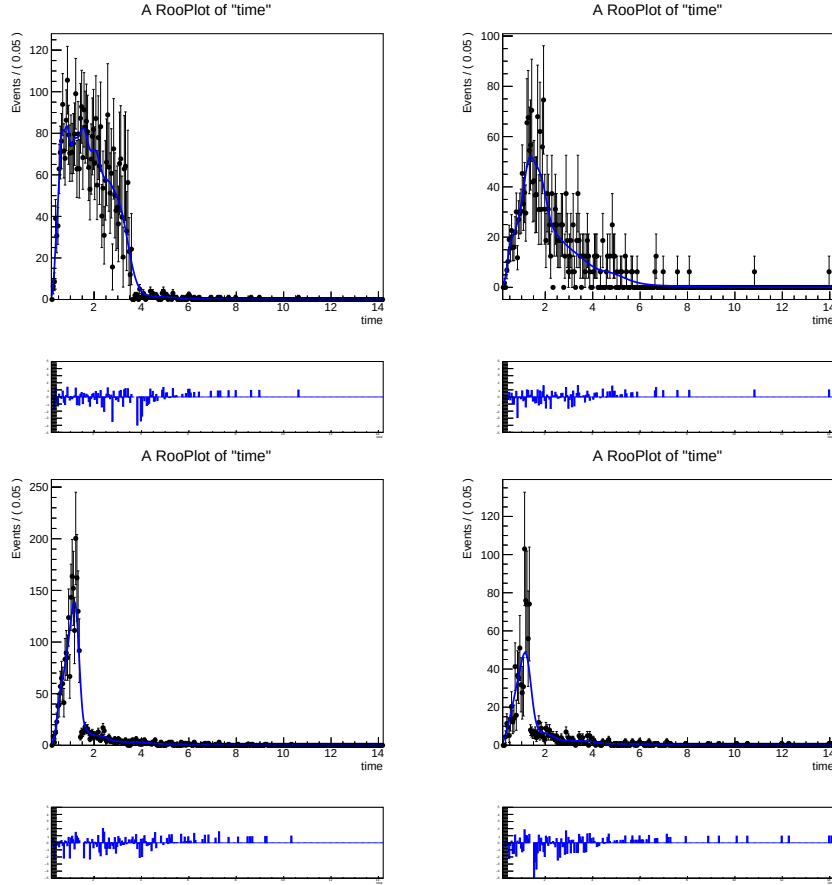


Figure 43: Template for combinatorial background. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

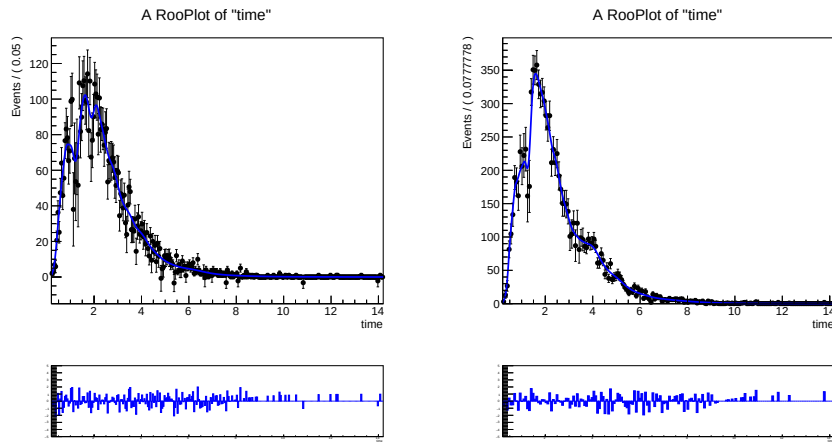


Figure 44: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KK.

5.1.3 Final state: KPI

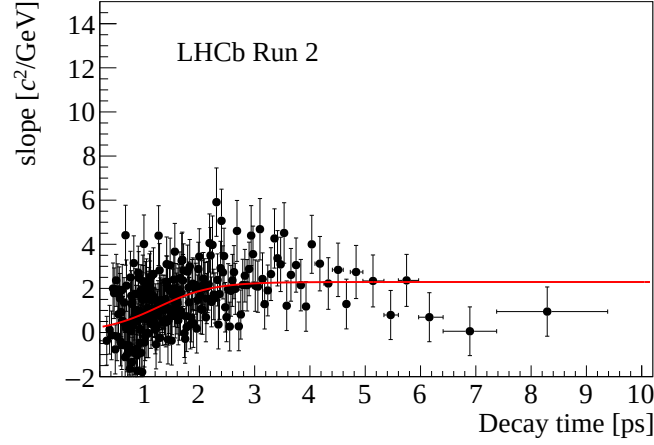


Figure 45: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging: None

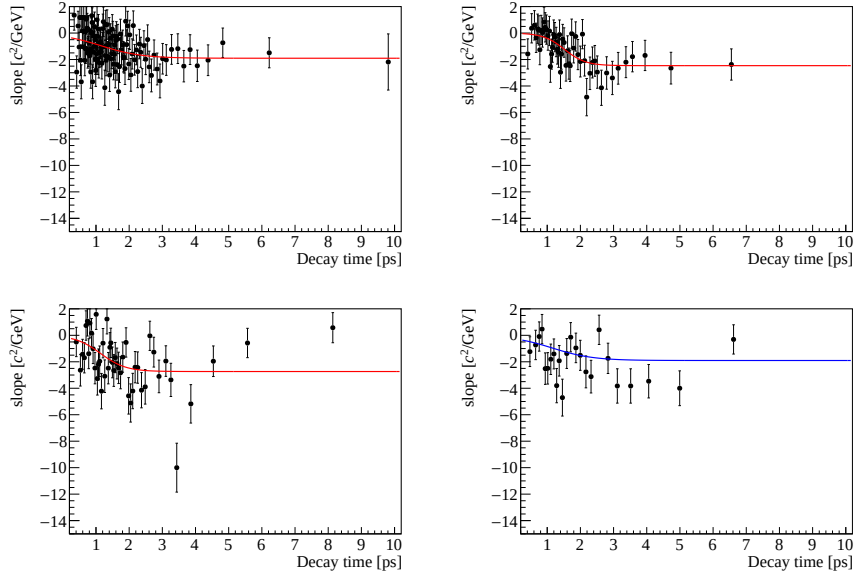


Figure 46: Invariant mass slope as a function of the decay time in the right-hand sideband ($m \in [5.5, 6.2]$ GeV). Year: Tot; Magnet: Tot; **bdtGraNEW** > 0.86; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

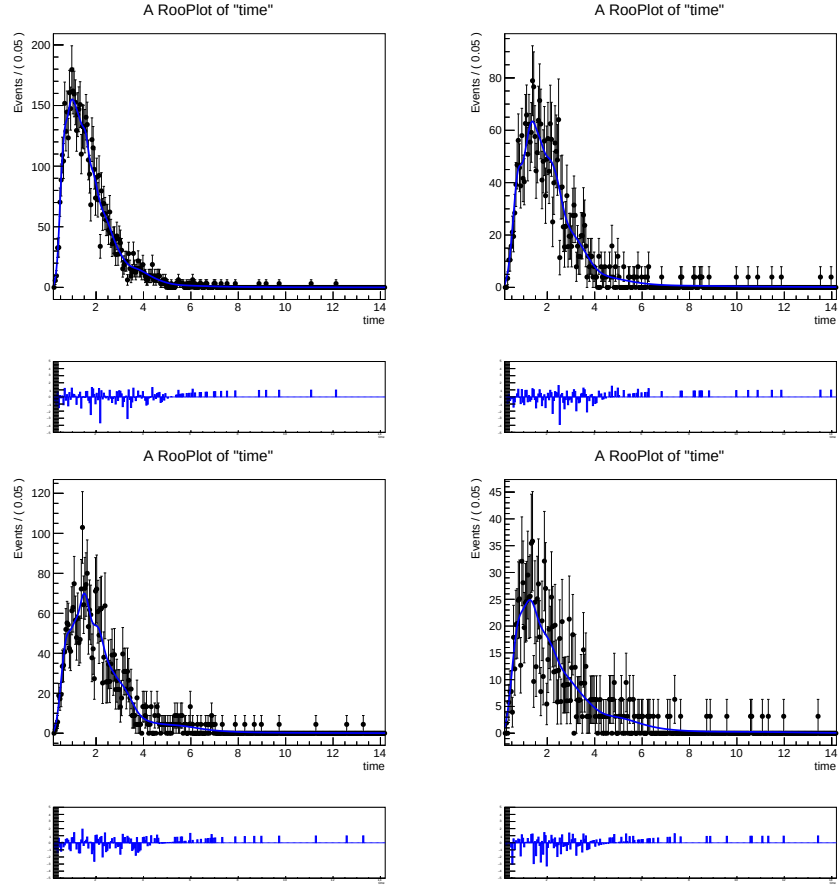


Figure 47: Template for combinatorial background. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI; F. Tagging (from top-left to bottom-right): OS and SS tagged, OS only tagged, SS only tagged, untagged.

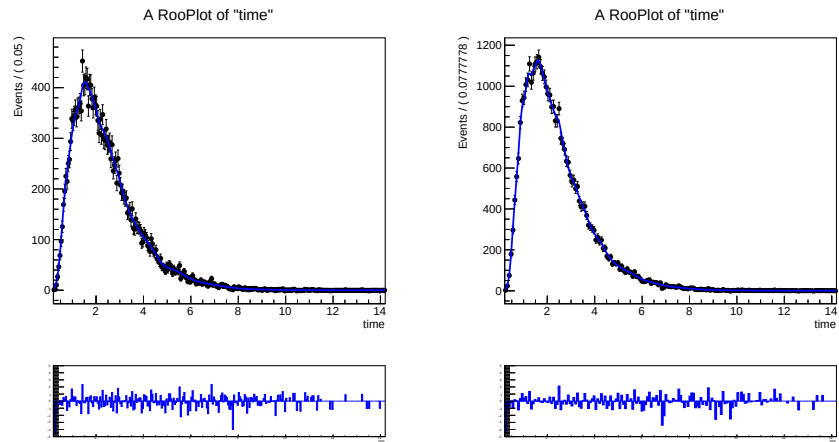


Figure 48: Templates for partially reconstructed bakground. Left: SS-untagged, right: SS-tagged. Year: Tot; Magnet: Tot; $\text{bdtGraNEW} > 0.86$; Spectrum: KPI.

6 Notes

nominal, with all years included. Ignoring of the SS tagger can be forced in `createTemplate.C`